Program : **Diploma in Biomedical Engineering**	
Course Code : **5241**	Course Title: **Medical Imaging Techniques**
Semester : **5**	Credits: **4**
Course Category: **Program Core**	
Periods per week: **4 (L:3, T:1, P:0)**	Periods per semester: **60**

Course Objectives:

- To lay foundation of various imaging modalities in the medical field.
- To impart knowledge on the principle of operation and instrumentation of various imaging techniques and its applications.

Course Prerequisites:

Topic	Course code	Course Title	Semester
Periodic Motion and Waves		Applied Physics I	1
Electromagnetism		Applied Physics II	2
Atomic structure		Applied Chemistry	1

Course Outcomes
On completion of the course, the student will be able to:

COn	Description	Duration (in hours)	Cognitive Level
CO1	Compare and contrast X-ray machine, CT machine and Fluoroscopy machine based on working principle and instrumentation.	20	Understanding
CO2	Describe the process of image formation in ultrasound machine and list the applications of ultrasound imaging.	11	Understanding
CO3	Identify the parts of the MRI machines and explain the role of each part.	16	Understanding
CO4	Explain nuclear imaging modalities and relate its applications in the medical field.	11	Understanding
	Series Test	2	

CO-PO Mapping:

Course Outcomes	PO1	PO2	PO3	PO4	PO5	PO6	PO7
CO1	2						
CO2	2						
CO3	2						
CO4	2						

3-Strongly mapped, 2-Moderately mapped, 1-Weakly mapped

Course Outline:

Module Outcomes	Description	Duration (Hours)	Cognitive Level
CO1	**Compare and contrast X-ray machine, CT machine and Fluoroscopy machine based on working principle and instrumentation.**		
M1.01	Explain the properties and generation of x-rays	3	Understanding
M1.02	Illustrate the working of x-ray machine and list the troubleshooting procedure of X-ray machine	5	Understanding
M1.03	Illustrate the working principle and block diagram of CT machine	8	Understanding
M1.04	Summarize the working of interventional radiological equipment	4	Understanding
Contents: **X-ray machine and CT machine** Definition of X- Rays, properties of X-ray, unit of x-ray, production of x-rays-principle of operation - basic x-ray tube - stationary anode x-ray tube - rotating anode x-ray tube, Block diagram of X-ray machine. Principle of Digital radiography. Computed tomography - Basic principle of computed tomography -Linear attenuation coefficient - CT number - concept of image reconstruction, Block diagram of CT machine- Scanning unit - CT generations - X-ray detectors(scintillation and xenon) - Processing system-viewing system, Interventional radiography – Fluoroscopy - block diagram of C-ARM, Applications of fluoroscopy - angiography, angioplasty. Case study : Prepare a presentation on the troubleshooting procedure of an x-ray machine			
CO2	**Describe the process of image formation in ultrasound machine and list the applications of ultrasound imaging.**		
M2.01	Describe the properties of ultrasound and relate it with principle of operation of ultrasound scanners.	4	Understanding
M2.02	Illustrate the modes of scanning used in ultrasound imaging	3	Understanding

<table>
<tr><td>M2.03</td><td>Describe the types of ultrasound scanners</td><td>3</td><td>Understanding</td></tr>
<tr><td>M2.04</td><td>List the applications of ultrasound scanning</td><td>1</td><td>Remembering</td></tr>
<tr><td></td><td>Series Test -I</td><td>1</td><td></td></tr>
<tr><td colspan="4">Contents:
Ultrasound Machine
Ultrasound - physics of ultrasound - properties of ultrasound-interaction of ultrasound with matter - acoustic impedance - Doppler effect.
Generation of ultrasound - Piezoelectric effect, Ultrasound transducer construction.
Modes of scanning - A-mode, B-mode & M-mode, Block diagram of A-mode and B-mode scanners.
Types of scanners - Linear scanner, Sector scanner, compound scanners & real time scanners (concept only).
Applications of Ultrasound - Doppler Ultrasound - Echocardiography – Echoencephalography - other applications in medical imaging.</td></tr>
<tr><td>CO3</td><td colspan="3">Identify the parts of the MRI machines and explain the role of each part.</td></tr>
<tr><td>M3.01</td><td>Describe the principle of operation of MRI</td><td>5</td><td>Understanding</td></tr>
<tr><td>M3.02</td><td>Explain the concept of relaxation times in MRI</td><td>2</td><td>Understanding</td></tr>
<tr><td>M3.03</td><td>Illustrate the block diagram of MRI machine and identify the parts of MRI machine</td><td>6</td><td>Understanding</td></tr>
<tr><td>M3.04</td><td>List the applications of MRI and explain the principle of fMRI</td><td>3</td><td>Understanding</td></tr>
<tr><td colspan="4">Contents:
MRI machine
Magnetic Resonance Imaging - Principle of operation - concept of magnetic dipole - precession - Lamour frequency - Concept of resonance- RF excitation - FID signal - Longitudinal and Transverse relaxation time - concept of image formation.
Block diagram of an MRI machine - Magnets used in MRI - RF coils - gradient coils - pulse sequence (basic spin-echo) - Functioning of each part.
Applications of MRI - cardiology, neurology, orthopedics etc
Functional MRI (fMRI) - working principle.</td></tr>
<tr><td>CO4</td><td colspan="3">Explain nuclear imaging modalities and relate its applications in the medical field.</td></tr>
<tr><td>M4.01</td><td>Explain the concept of nuclear imaging and radioactive decay</td><td>3</td><td>Understanding</td></tr>
<tr><td>M4.02</td><td>Describe the working of Gamma camera.</td><td>2</td><td>Understanding</td></tr>
<tr><td>M4.03</td><td>Describe the principle of operation and block diagram of SPECT scanner</td><td>3</td><td>Understanding</td></tr>
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M4.04	Describe the principle of operation and block diagram of PET scanner	3	Understanding
	Series Test -II	1	

Contents:

Nuclear imaging

Concept of Nuclear imaging- Radioactive decay-Half life period - properties of Gamma rays.

Gamma camera - working principle- block diagram- gamma detectors - pulse height analyzer - Applications of gamma camera - isotopes used,

Emission Computed Tomography- PET scanner - Principle of operation -Annihilation - block diagram - Applications - isotopes used.

SPECT scanners - Principle of operation -block diagram- Application - isotopes used.

Case study: Prepare a video on operation of a Gamma camera and radiopharmaceutical preparation.

Text / Reference:

T/R	Book Title/Author
T1	RS Khandpur, *Handbook of Biomedical Instrumentation*- 3rd Edition, Tata McGraw-Hill Pub. India, 2014
T2	Steve Webb, *Physics of Medical Imaging*- 2nd Edition, Institute of Physics Publishing, U K, 2008
R1	Massey & Meredith ,*Fundamentals Physics of radiology*- J. Wright; 2nd edition - 1972
R2	A C Kak, *Principles of computerized tomographic imaging,*1st edition, - Society for Industrial and Applied Mathematics, Philadelphia, 2001
R3	Govind B Chavhan, *MRI Made Easy (for Beginners),* 1st edition, Jaypee Brothers Medical Publishers Pvt. Ltd, India, 2013

Online resources:

Sl. No	Website Link
1	https://www.youtube.com/watch?v=whp2pnCBs6s
2	www.khanacademy.org
3	https://www.youtube.com/watch?v=DSX0ygI0fZs
4	https://www.wikihow.com
5	https://www.youtube.com/watch?v=IAWnRnqpVP0
6	https://www.youtube.com/watch?v=bXxqGWj2340
7	https://www.youtube.com/watch?v=NKVyHFk01lY
8	https://www-pub.iaea.org/books/iaeabooks/8841/diagnostic-radiology-physics-a-handbook-for-teachers-and-students